

Purpose Overview: Griffiths Electrodynamics & Schroeder Thermal Physics

Electrodynamics by Griffiths

Chapter	Topic	Purpose
1	Vector Analysis	Build mathematical language required to describe fields precisely
2	Electrostatics	Establish fundamental laws governing stationary charges and fields
3	Special Techniques	Develop advanced methods to solve otherwise intractable electrostatic problems
4	Electric Fields in Matter	Understand how materials respond to electric fields at a macroscopic level
5	Magnetostatics	Introduce magnetic fields from steady currents and their governing laws
6	Magnetic Fields in Matter	Explain magnetization and material response to magnetic fields
7	Electrodynamics	Unify electricity and magnetism into time-dependent Maxwell framework
8	Conservation Laws	Connect EM theory with energy and momentum conservation principles
9	Potentials and Fields	Provide deeper formulation using potentials and causality (retardation)
10	Radiation	Explain how accelerating charges emit electromagnetic waves

Thermal Physics by Schroeder

Chapter	Topic	Purpose
1	Energy in Thermal Physics	Introduce thermodynamic quantities and energy flow concepts
2	Second Law	Formalize irreversibility and entropy as central principles
3	Interactions and Implications	Apply entropy concepts to real physical processes
4	Engines and Refrigerators	Analyze practical systems governed by thermodynamic laws
5	Free Energy	Introduce criteria for equilibrium and spontaneous processes
6	Boltzmann Statistics	Link microscopic states to macroscopic entropy
7	Classical Statistical Mechanics	Provide tools to compute thermodynamic quantities from first principles
8	Quantum Statistical Mechanics	Extend statistical framework to quantum systems
9	Applications	Apply statistical mechanics to real systems like solids and phase changes